Computational Anatomy of Kimi K3

Deep Dive into Kimi K3 — Chapter 2

Yin Yang

Foundation Books Project

The one idea of this chapter

K3 keeps Chapter 1's prediction task and its three mixing jobs, and changes *how* sequence, channel, and depth information are combined.

The one idea of this chapter

K3 keeps Chapter 1's prediction task and its three mixing jobs,
and changes *how* sequence, channel, and depth information
are combined.

Goal: name each new mechanism by the job it does, then follow a picture question through prefill and one continuation call — and see exactly which state crosses the call boundary.

Roadmap

Three jobs, new mechanisms

Sequence: two memories of one prefix

Channel: selecting subnetworks

Depth: selecting earlier outputs

The picture question, end to end

What grows, and the route ahead

Three jobs, new mechanisms

Three questions from the baseline decoder

sequence
KDA + gated MLA

channel
shared +
routed experts

depth
Block AttnRes

Three questions from the baseline decoder

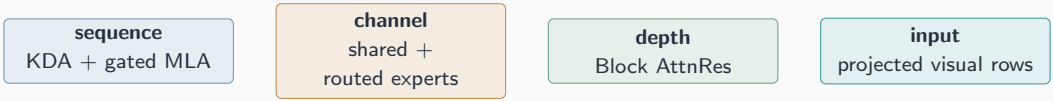
sequence
KDA + gated MLA

channel
shared +
routed experts

depth
Block AttnRes

input
projected visual rows

Three questions from the baseline decoder



Running request: a red circle above a blue square — "Which shape is below the red circle?" — with `square` stipulated as the answer token.

Sequence: two memories of one prefix

Fixed-extent state versus compact records

KDA: $S_t \rightarrow S_{t+1}$
same shape, new contents

Fixed-extent state versus compact records

KDA: $S_t \rightarrow S_{t+1}$
same shape, new contents



MLA: one compact record appended per position

Fixed-extent state versus compact records



Recurrent state carries an accumulated account; records let a later query weight separate positions. K3 interleaves three KDA layers with one gated MLA layer, so total retained memory still grows — through MLA.

Channel: selecting subnetworks

Router at full width, experts in a latent

- Four routed experts, two selected, one shared always runs.

Router at full width, experts in a latent

- Four routed experts, two selected, one shared always runs.
- The router reads the 8-feature row; selected experts read a 4-feature latent; the result is normalized, projected back to 8, and added to the shared path.

Router at full width, experts in a latent

- Four routed experts, two selected, one shared always runs.
- The router reads the 8-feature row; selected experts read a 4-feature latent; the result is normalized, projected back to 8, and added to the shared path.

The route decides which networks run; the projection decides what they process. Three evaluations is not an active-parameter fraction.

Depth: selecting earlier outputs

Same position, different depths

$$0.2 \underbrace{(1)}_{x_0} + 0.5 \underbrace{(5)}_{b_1} + 0.3 \underbrace{(4)}_{v_3} = 3.9$$

- Sources: embedding, completed block summaries, the partial sum — all the *same token* at different depths.

Same position, different depths

$$0.2 \underbrace{(1)}_{x_0} + 0.5 \underbrace{(5)}_{b_1} + 0.3 \underbrace{(4)}_{v_3} = 3.9$$

- Sources: embedding, completed block summaries, the partial sum — all the *same token* at different depths.
- The workspace ends with the call; it is never a cross-call token cache.

Same position, different depths

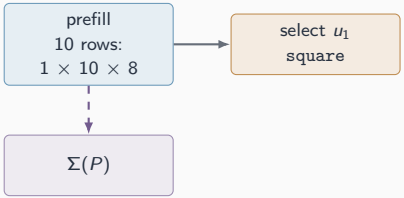
$$0.2 \underbrace{(1)}_{x_0} + 0.5 \underbrace{(5)}_{b_1} + 0.3 \underbrace{(4)}_{v_3} = 3.9$$

- Sources: embedding, completed block summaries, the partial sum — all the *same token* at different depths.
- The workspace ends with the call; it is never a cross-call token cache.

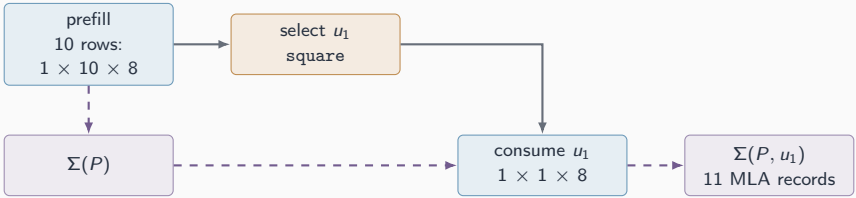
Depth selection precedes each sequence and each channel transformation; only the sequence transformation updates retained state.

The picture question, end to end

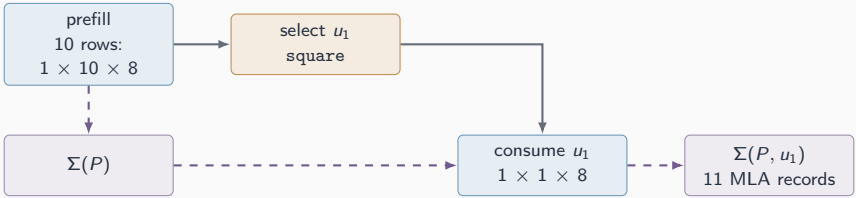
Selection is not consumption



Selection is not consumption



Selection is not consumption



Joint hybrid-prefix consistency: KDA states and MLA records must describe the same processed prompt and accepted suffix — routes and depth workspaces never cross the call.

What grows, and the route ahead

One coordinate at a time, then the specialist chapters

- Longer prefix: MLA appends; KDA changes contents at fixed extent.
- Bigger pool E at fixed k: more stored experts, same selected count.

One coordinate at a time, then the specialist chapters

- Longer prefix: MLA appends; KDA changes contents at fixed extent.
- Bigger pool E at fixed k: more stored experts, same selected count.
- Logical deductions — not measured bytes, quality, or latency.

One coordinate at a time, then the specialist chapters

- Longer prefix: MLA appends; KDA changes contents at fixed extent.
- Bigger pool E at fixed k: more stored experts, same selected count.
- Logical deductions — not measured bytes, quality, or latency.

Next: Chapter 3 (sparse MoE), Chapters 4–6 (KDA, hybrid, gated MLA), 7–8 (AttnRes, Stable LatentMoE), 9 (vision), and Chapter 10's integrated K3.